STOR 557 HW 1

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Question 1

```
nukes=
  read.table('https://rls.sites.oasis.unc.edu/faculty/rs/source/Data/nukes.dat',header=T)

nukes$LC = log(nukes$C)
nukes$LT1 = log(nukes$T1)
nukes$LT2 = log(nukes$T2)
nukes$LS = log(nukes$S)
nukes$LN = log(nukes$N)

m1 = lm(LC ~ D + LT1 + LT2 + LS + PR + NE + CT + BW + LN + PT, data=nukes)
summary(m1)
```

```
##
## Call:
## lm(formula = LC ~ D + LT1 + LT2 + LS + PR + NE + CT + BW + LN +
##     PT, data = nukes)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.28574 -0.10408  0.02784  0.09512  0.25031
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -14.24198    4.22880  -3.368  0.00291 **
## D             0.20922    0.06526   3.206  0.00425 **
## LT1           0.09187    0.24396   0.377  0.71025
## LT2           0.28553    0.27289   1.046  0.30731
## LS            0.69373    0.13605   5.099 4.75e-05 ***
## PR           -0.09237    0.07730  -1.195  0.24542
## NE            0.25807    0.07693   3.355  0.00300 **
## CT            0.12040    0.06632   1.815  0.08376 .
## BW            0.03303    0.10112   0.327  0.74715
## LN           -0.08020    0.04596  -1.745  0.09562 .
## PT           -0.22429    0.12246  -1.832  0.08125 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1645 on 21 degrees of freedom
```

```
## Multiple R-squared:  0.8717, Adjusted R-squared:  0.8106
## F-statistic: 14.27 on 10 and 21 DF,  p-value: 3.081e-07
```

```
m2 = lm(LC ~ D + LT1 + LT2 + LS + PR + NE + CT + LN + PT, data=nukes)
m3 = lm(LC ~ D + LT2 + LS + PR + NE + CT + LN + PT, data=nukes)
m4 = lm(LC ~ D + LS + PR + NE + CT + LN + PT, data=nukes)
m5 = lm(LC ~ D + LS + NE + CT + LN + PT, data=nukes)
summary(m5)
```

```
##
## Call:
## lm(formula = LC ~ D + LS + NE + CT + LN + PT, data = nukes)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.32721 -0.07620  0.02920  0.08115  0.28946
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -13.26031    3.13950  -4.224 0.000278 ***
## D             0.21241    0.04326   4.910 4.70e-05 ***
## LS            0.72341    0.11882   6.088 2.31e-06 ***
## NE            0.24902    0.07414   3.359 0.002510 **
## CT            0.14039    0.06042   2.323 0.028582 *
## LN           -0.08758    0.04147  -2.112 0.044891 *
## PT           -0.22610    0.11355  -1.991 0.057490 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1592 on 25 degrees of freedom
## Multiple R-squared:  0.8569, Adjusted R-squared:  0.8225
## F-statistic: 24.95 on 6 and 25 DF,  p-value: 2.058e-09
```

```
m6 = lm(LC ~ D + LS + NE + CT + LN, data=nukes)
AIC(m1, m2, m3, m4, m5, m6)
```

```
##    df       AIC
## m1 12 -14.19023
## m2 11 -16.02804
## m3 10 -17.91154
## m4  9 -17.56781
## m5  8 -18.68713
## m6  7 -15.97632
```

Based on the question I transformed the variables C, T1, T2, S, and N using log. After that I made the full regression model first then I used backwards elimination in order to find the "best" model. After every model I would drop the variable that had the highest P value until I got to the point where all the variables would be statistically significant. I found that model 5 which included D, LS, NE, CT, LN, and PT was the best out of the 5 that I tested since it had the the second highest adjusted R^2 only by 0.0031 but as well as the lowest AIC at -18.68.

Question 2

```
newplant = data.frame(
  D = 67.99,
  LS = log(1065),
  NE = 0,
  CT = 0,
  LN = log(8),
  PT = 0
)

exp(predict(m5, newdata = newplant, interval = "confidence", level = 0.95))
```

```
##        fit      lwr      upr
## 1 420.8708 356.9322 496.2629
```

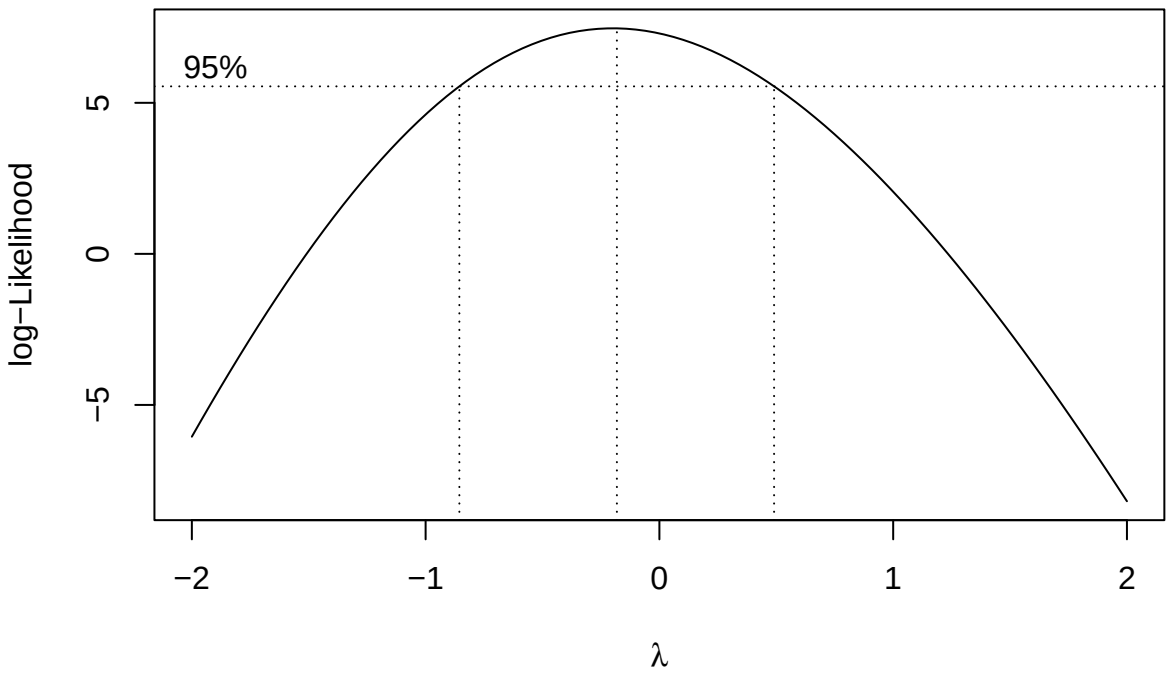
```
exp(predict(m5, newdata = newplant, interval = "prediction", level = 0.95))
```

```
##        fit      lwr      upr
## 1 420.8708 291.5877 607.4749
```

The 95% confidence interval is (356.932, 496.3639) which means that the estimated mean cost for the plant with these certain characteristics is between $356.93 million to $496.26 million. The 95% prediction interval is (291.5877, 607.4739) which estimates the possible cost of one individual new plant itself to be between $291.59 million to $607.47 million.

Question 3

```
m5_original = lm(C ~ D + LS + NE + CT + LN + PT, data=nukes)
library(MASS)
boxcox(m5_original)
```




```
m_identity = lm(C ~ D + LS + NE + CT + LN + PT, data=nukes)

m_sqrt = lm(sqrt(C) ~ D + LS + NE + CT + LN + PT, data=nukes)

m_log = lm(LC ~ D + LS + NE + CT + LN + PT, data=nukes)

summary(m_identity)$adj.r.squared
```

```
## [1] 0.7737515
```

```
summary(m_sqrt)$adj.r.squared
```

```
## [1] 0.8032976
```

```
summary(m_log)$adj.r.squared
```

```
## [1] 0.8225321
```

```
m_Soriginal = lm(LC ~ D + S + NE + CT + LN + PT, data=nukes)
summary(m_Soriginal)$adj.r.squared
```

```
## [1] 0.8151798
```

```r
summary(m_log)$adj.r.squared
```

```
## [1] 0.8225321
```

```r
m_Noriginal = lm(LC ~ D + LS + NE + CT + N + PT, data=nukes)
summary(m_Noriginal)$adj.r.squared
```

```
## [1] 0.8108462
```

```r
summary(m_log)$adj.r.squared
```

```
## [1] 0.8225321
```

Even though our maximum is around lambda = -0.2, our lambda = 0 is contained within our 95% confidence interval. This means using Log(C) would be appropriate for our models. I tested other transformations of C including the original, square root, and log and they all have relatively similar adjusted R^2 values. Since only LS and LN were in my final model, I also tested it with the original S and N and found that Log(S) had a higher adjusted R^2 while the original N had a higher adjusted R^2.